

# Introduction to remote sensing

## Animal tracking 25/26

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# A brief introduction on Remote sensing

- ▶ These slides have been adapted from the presentation by Ines Standfuß for the AniMove symposium
- ▶ The purpose of this presentation is to briefly introduce you to the principles of remote sensing, a very important discipline that allows the collection of environmental data

- ▶ Earth observation (EO) is the act of gathering of information about Earth. It encompasses numerous techniques and approaches, ONE of which is remote sensing
- ▶ Remote sensing (RS) is the science of obtaining information about objects without being in direct physical contact with the object. It Under the umbrella of RS we have earth observation from satellites as well as from drones/aircrafts

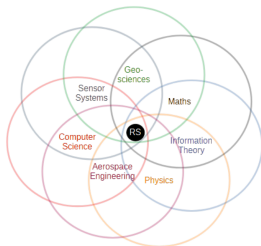


Figure 1: RS is a multidisciplinary topic and there is no know-it-all expert

- ▶ RS typically sense electromagnetic radiations (light and/or radio waves), and these are the two kinds of sensed radiations we will cover throughout the presentation as they are the two of most interest for biological and conservation work
- ▶ Electromagnetic radiation is a synchronized oscillation of electric and magnetic fields!
- ▶ It can be seen as particles (photons) behaving like a wave (i.e., having a wavelength and a frequency
  - ▶ wavelength ( $\lambda$ ): distance covered by a full oscillation
  - ▶ frequency ( $\nu$ ): speed of a full oscillation (photons travel, in a vacuum, at the speed of light: 300.000 Km/s. Their frequency is the speed of light divided by the wavelength)

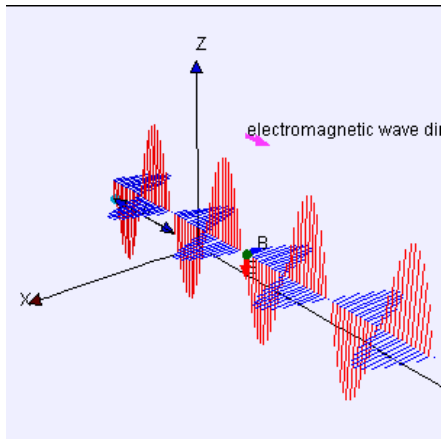


Figure 2: A graphical representation of a self-propagating electromagnetic field (image [source](#))

- ▶ The wavelength (or the frequency, as they can be used almost invariably) is a core characteristic of the measurement
- ▶ Visible' light (VIS): wavelengths ranging from roughly 400 to 700nm
- ▶ Infra-Red (IR): longer wavelengths from about 700nm up to a few hundred micro-meters
  - ▶ Near Infra-Red (NIR): 700-1,000nm is just the same as the visible light (but we don't see it)
  - ▶ Short Wave-Infra-Red (SWIR): 1,000-2,500nm is quite similar in behaviour
  - ▶ Mid-Wave Infra-Red (MWIR): 3,000 – 6,000nm and Long-Wave Infra-Red (LWIR): 7,500-14,000 nm corresponds to thermal radiations

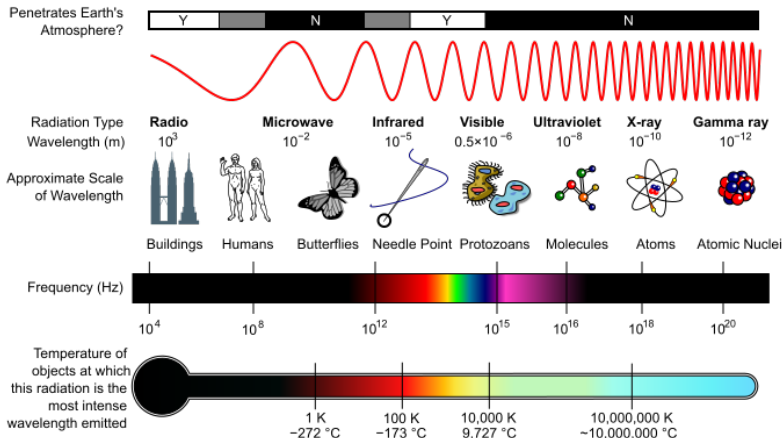
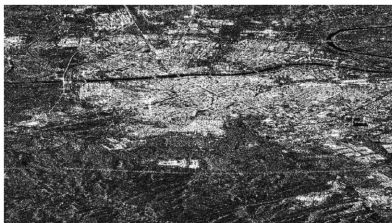


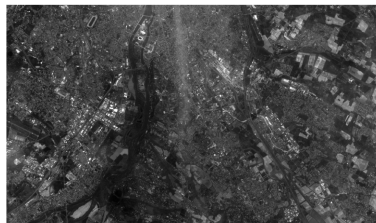
Figure 3: A diagram of the electromagnetic spectrum, showing various properties across the range of frequencies and wavelengths ([Image source](#))

- ▶ The way radiations interact with the matter depends on their wavelength
- ▶ To measure different kind of radiations we need to use different types of sensors
  - ▶ Sensors measuring visible to SWIR can be seen as very high end digital cameras
  - ▶ To measure microwaves and above we need to use antennas
- ▶ What is exactly measured also depends on the wavelength
  - ▶ In VIS we mostly measure solar radiation
  - ▶ Other domains of the spectrum measure different things





Sentinel 1



Sentinel 2

Figure 4: Radar (Sentinel 1) and multispectrum (Sentinel 2) image comparison. The area depicted is the same, but the type of image is very different and so is the information in it

- ▶ Electromagnetic radiations interact with the medium they encounter as a function of the wavelength
- ▶ Typically, as combination of absorption, reflection/scattering and transmission
- ▶ Example VIS domain:
  - ▶ A mirror is mostly reflecting
  - ▶ A window is mostly transmitting
  - ▶ A black solid is mostly absorbing

- ▶ These interactions bring along advantages and disadvantages:
  - ▶ Advantage: You can derive information about the medium composition
    - ▶ E.g. some gases absorb specific wavelengths making the atmosphere 'opaque' – characterising 'how opaque' they are, allows to infer the atmospheric composition
  - ▶ Disadvantage: You have to 'correct' for unwanted effects
    - ▶ E.g. before deriving information about the earth surface, you will have to correct for effects introduced by the atmosphere



Figure 5: The persons are only detectable in SWIR and IR images in presence of artificial fog (2). Moreover, if the artificial fog is too dense (3), the two persons in the background are only detectable in IR image (3). (Image [source](#))

|            | Passive                                                        | Active                                               |
|------------|----------------------------------------------------------------|------------------------------------------------------|
| Principles | Measures whatever comes to you                                 | Emits a radiation and measures what comes back       |
| Pros/Cons  | Lower energy consumption/Cannot measure if there is no energy! | High energy consumption/Better control of the signal |
| Example    | Digital camera, optical sensors                                | LiDAR, Radar, Digital cameras with flash             |

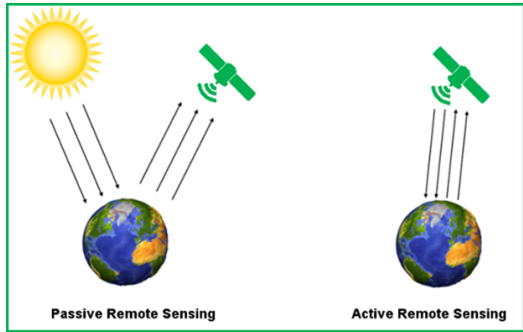


Figure 6: Schematic of the difference between active and passive energy source. Image [source](#)

- ▶ Most surfaces emit radiation depending on their temperature
- ▶ The intensity and wavelength depends on the temperature
- ▶ From 400 to 2,500 nm the main contributor is primarily the Sun (6,000 K)
- ▶ For MWIR to LWIR contributions from objects on Earth start being also observable
- ▶ These temperatures on Earth can reach a few 100°Celsius (e.g. Lava 1,200°C)
- ▶ Thus the (improper) name 'thermal' remote sensing

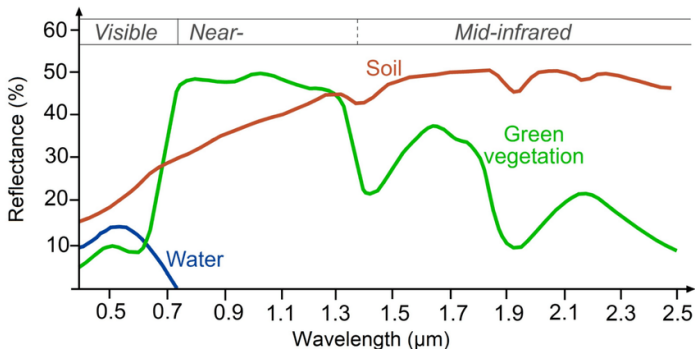


Figure 7: Typical spectral signatures of specific land cover types in the VIS and IR region of the electromagnetic spectrum. Adapted from Zanchetta (2017)



- ▶ In the VIS to SWIR the atmosphere is significantly interacting with Electro-magnetic radiations
- ▶ One example is the presence of clouds. About 67% of Earth's surface is covered by clouds at a given time (King, Platnick, Menzel, Ackerman, & Hubanks, 2013). It is slightly better over land, and slightly worse over waters
- ▶ Another example is the scattering of the 'shorter' wavelength. Molecules and aerosols sometime re-emit/ redirect radiations in a random direction. This typically affects 'smaller' radiation (in VIS - blue). This is the reason why the sky appears 'blue'
- ▶ You can't afford to 'ignore' the interaction with the atmosphere!!!

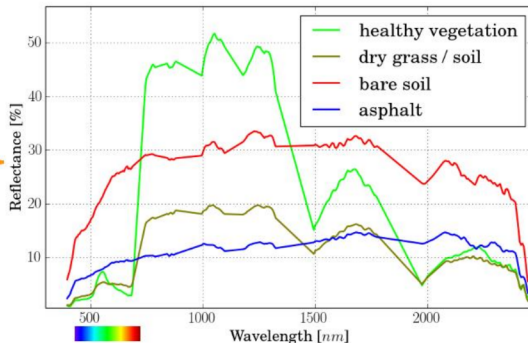


Figure 8: Inferring features from an image.

- King, M. D., Platnick, S., Menzel, W. P., Ackerman, S. A., & Hubanks, P. A. (2013). Spatial and temporal distribution of clouds observed by MODIS onboard the terra and aqua satellites. *IEEE Transactions on Geoscience and Remote Sensing*, 51(7), 3826–3852. Institute of Electrical; Electronics Engineers (IEEE).
- Zanchetta, A. (2017). Remote sensing techniques for change detection analysis in arid and semi-arid areas. [amsdottorato](#).